Bivariate Regression I

October 26, 2022

As you come in...

Load in the Fariss Human Rights data:

Data< −read.csv("https://raw.githubusercontent.com/Brandon-Bolte/PLSC309$_F$22/*main*/*Fariss*2010.*csv*")

Then, find and interpret the correlation between:

1. the number of treaties signed (TreatyCount) and a country's democracy score (polity2)
2. the country's Polity score (polity2) and its level of respect for human rights (respectHR)
3. the number of treaties signed (TreatyCount) and respect for human rights (respectHR)

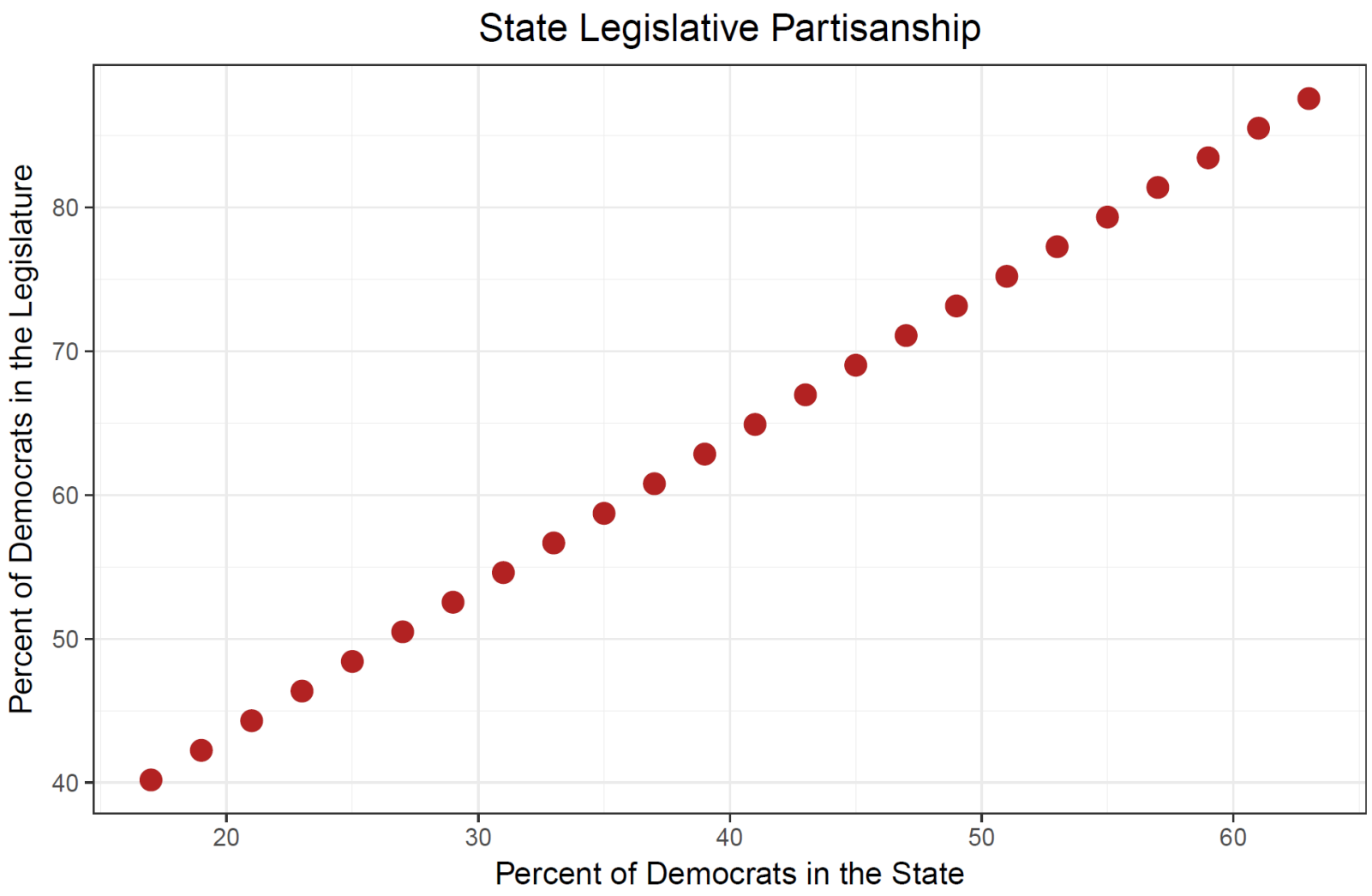
Are these correlations statistically significant? What do these results mean?

Bivariate Hypothesis Testing

	Independent Variable		
Dependent Variable		*Categorical*	*Continuous*
	Categorical	Crosstab/Chi-square	Logit/Probit
	Continuous	Difference of Means/T-test; Bivariate Regression	Correlation Coefficient; Bivariate Regression

A Question

How would we know that the percentage of Democrats in a state is related to the percentage of Democrats in a state legislature?


State Legislative Partisanship
Percent of Democrats in the Legislature
80
70
60
50
40
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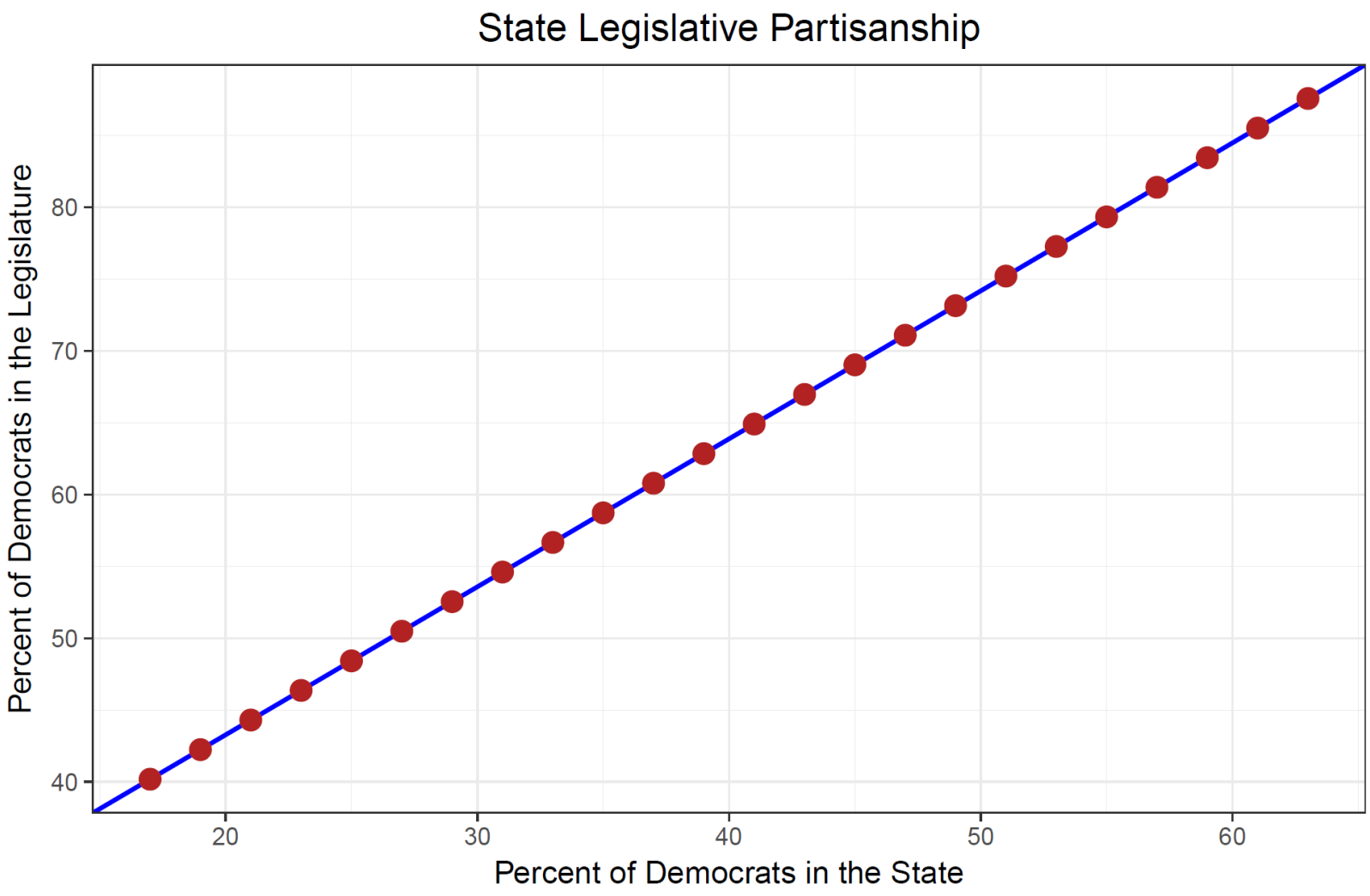
Percent of Democrats in the State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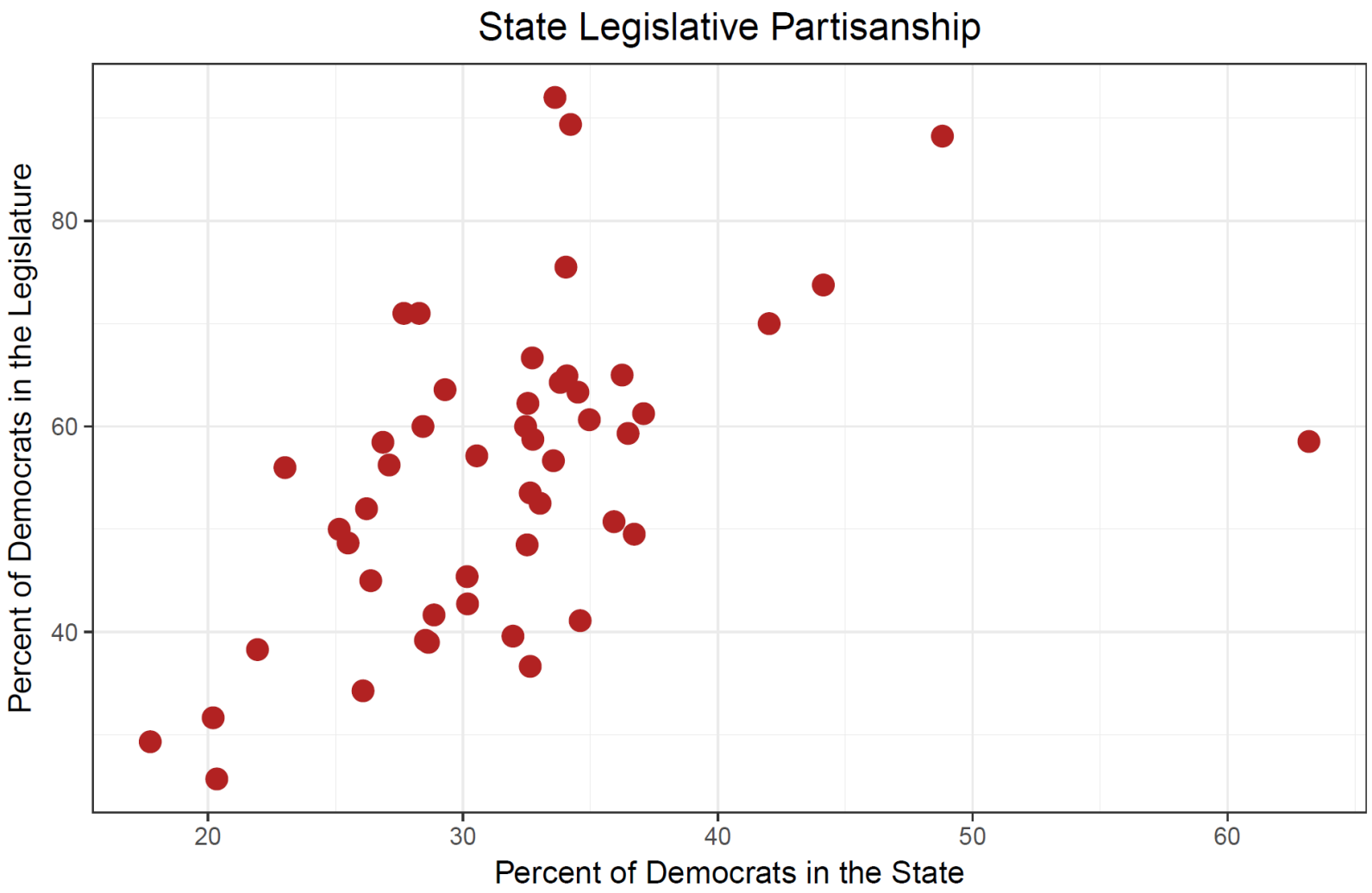
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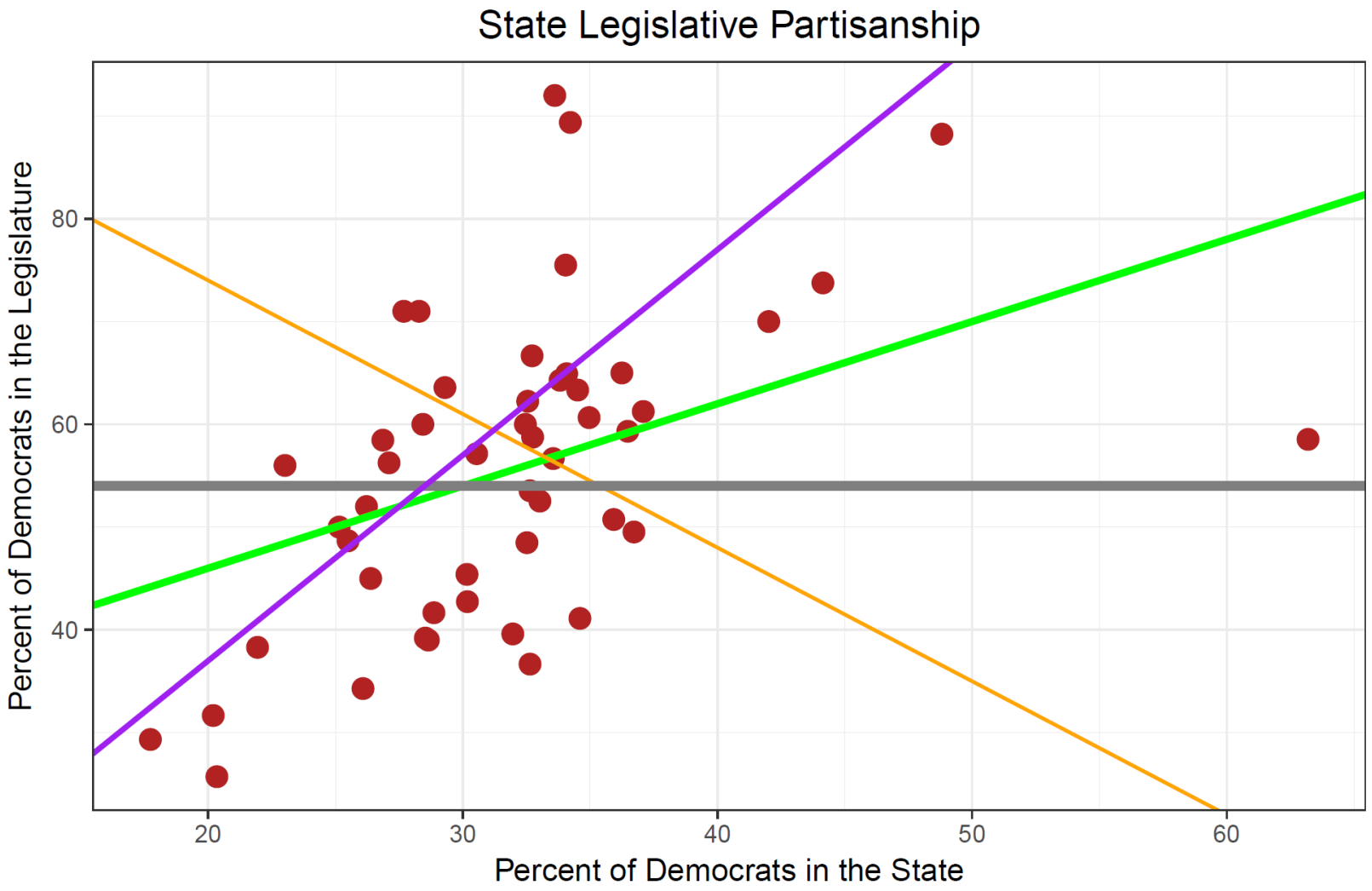
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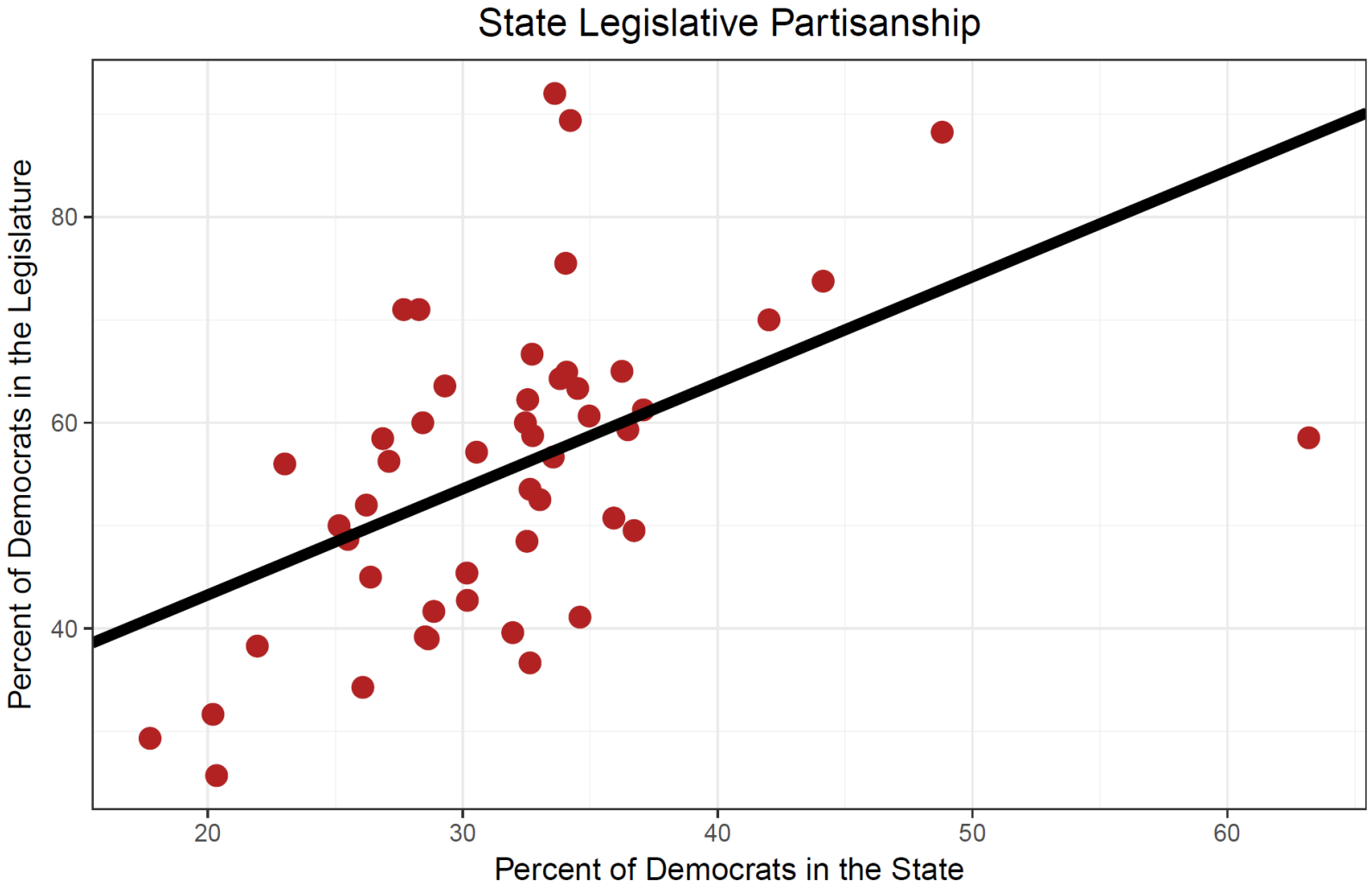
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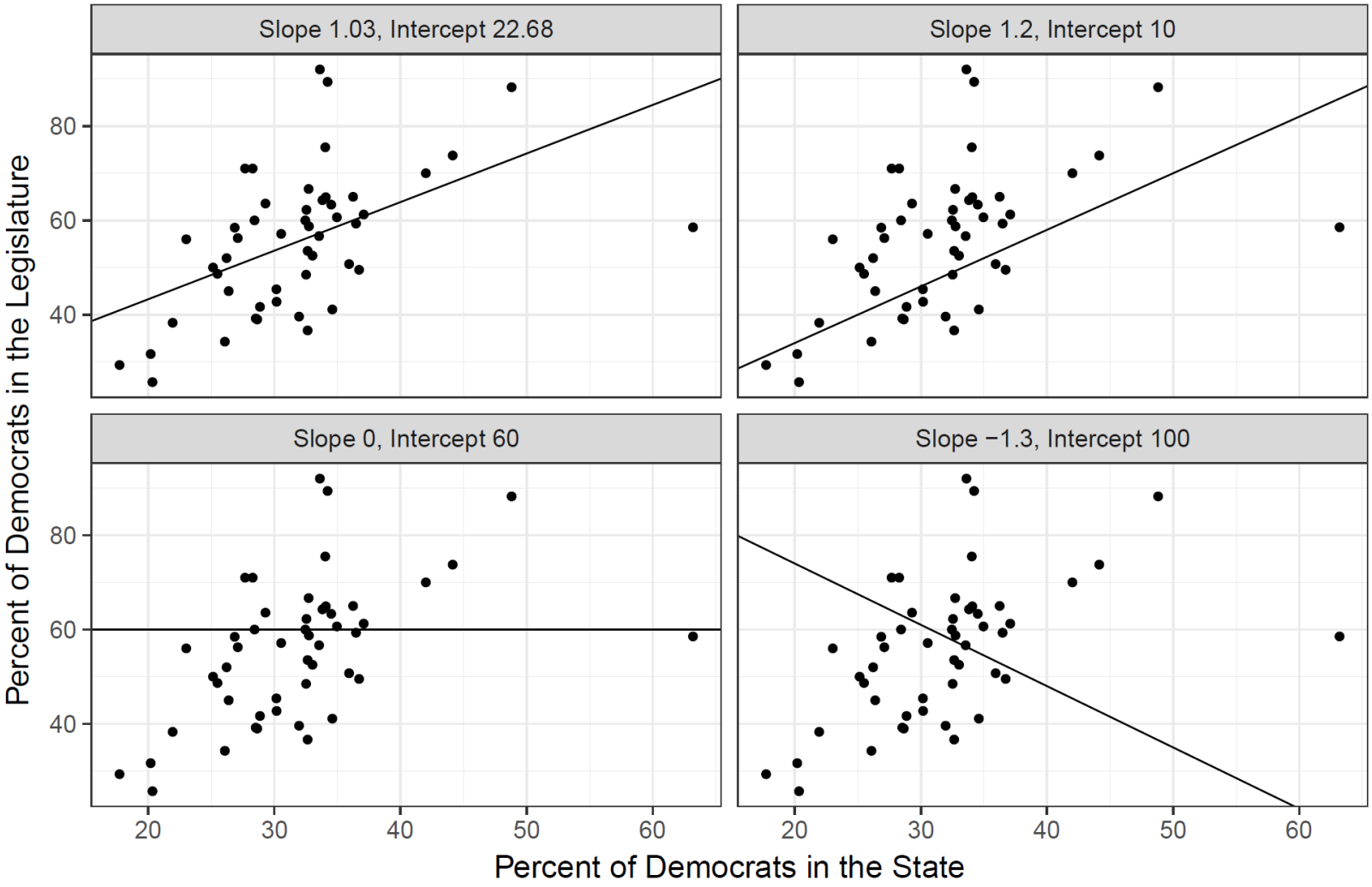
What is Regression?

Procedure: Drawing the best fitting straight line through a cloud of data

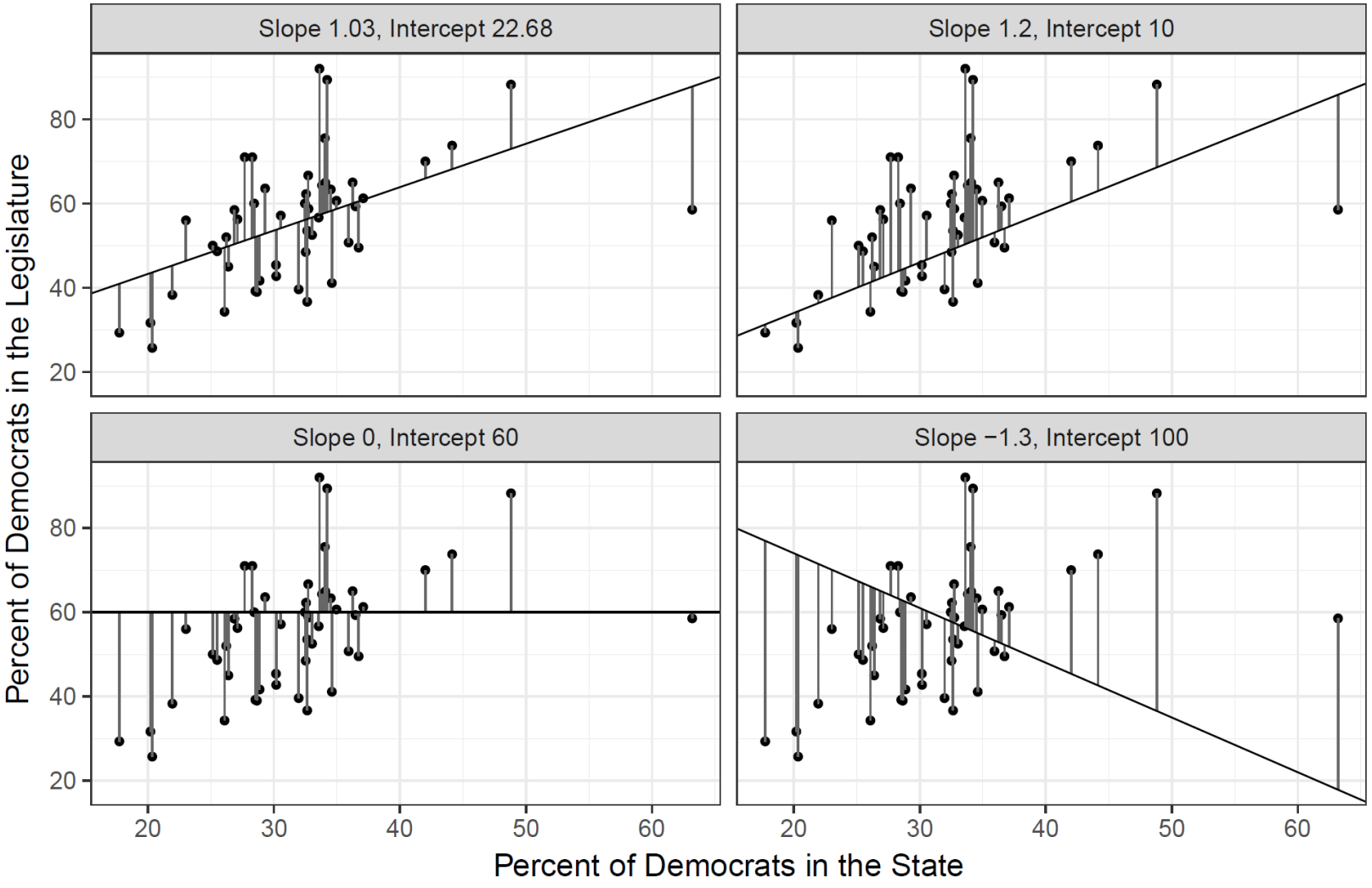
Why?

- Make an inference about the relationship between a dependent variable (Y) and one or more independent variables (X)

How do we choose?



We minimize the "errors"



The Equation for a Line

$$Y = a + bX \tag{1}$$

- Y = dependent variable (y axis)
- X = independent variable (x axis)
- a = intercept (value of Y when X = 0)
- b = slope (change in Y when X changes by 1 unit)

$$b = \frac{rise}{run}$$

Predicting Using Regression

$$Y = a + bX$$

$$\% \text{ Dems in Leg.} = 22.68 + (1.03 \times \% \text{ Dems in State})$$

- Calculate expected % Dem Legislators when there are 25% Dems in the state:
  - $22.68 + (1.03 \times 25) = 48.43$

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Note: What is the difference between our predicted Y values? **1.03**

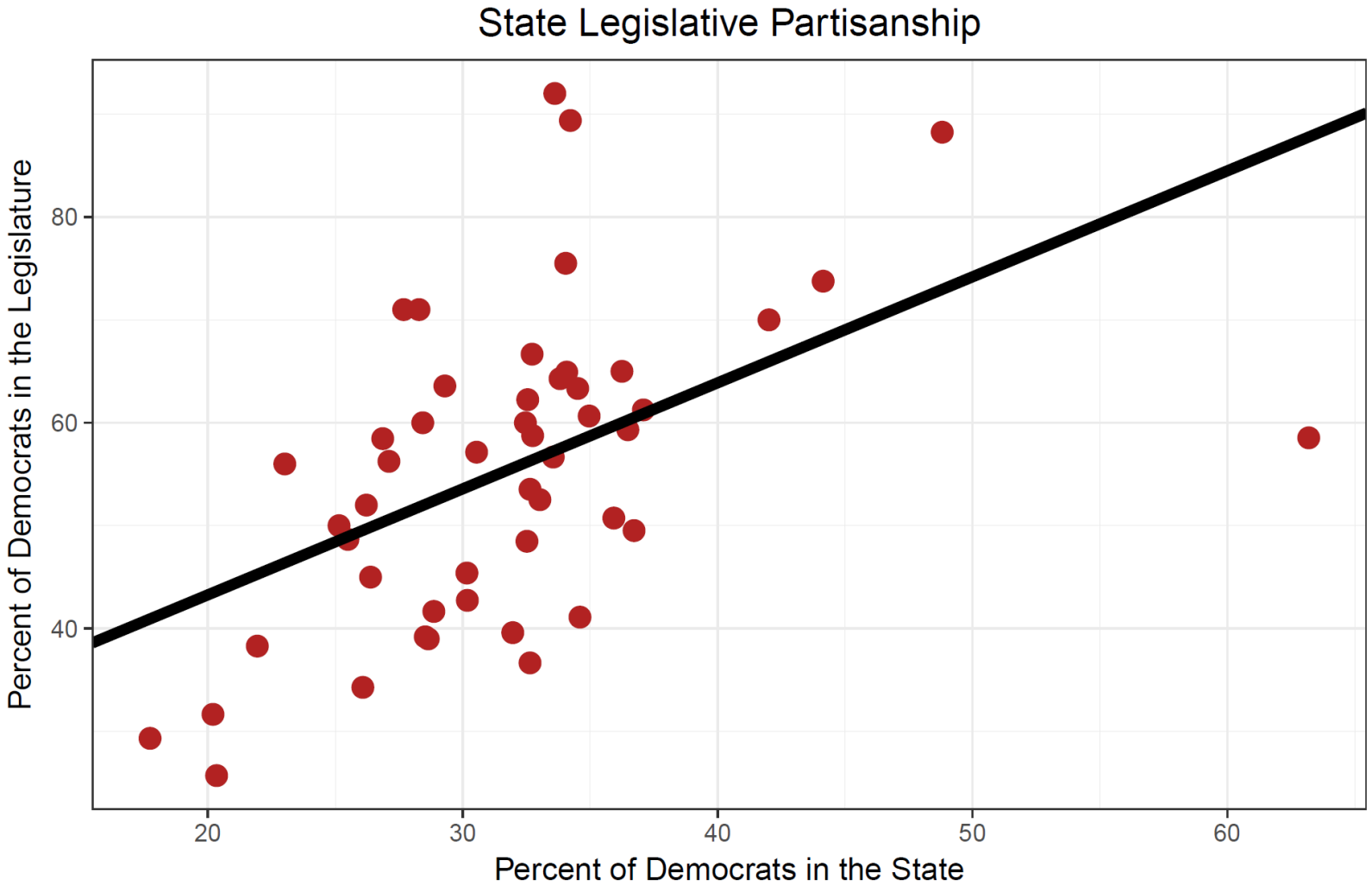
Inference from Regression

Beyond "prediction," the linear regression model can be used to perform inference about the relationship between two variables:

- Y is the DV
- X is the IV
- We are interested in the population relationship:

$$\mathbf{Y} = \alpha + \beta \mathbf{X}$$

  - $\mathbf{Y}$ = dependent variable (y axis)
  - $\mathbf{X}$ = independent variable (x axis)
  - α = intercept (value of Y when X = 0)
  - β = slope (change in Y when X changes by 1 unit)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Estimating Linear Regression Parameters

Since we are estimating a population regression line from a sample, and because there is always randomness in the world, our regression line equation also must include these random "errors," (residuals)

We assume that the distribution of the residuals is Normal (bell-shaped) with a mean of zero: $\epsilon_i \sim \mathcal{N}(0, \sigma^2)$

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The estimated linear regression equation is therefore:

$$\boldsymbol{Y_i} = \alpha + \beta \boldsymbol{X_i} + \epsilon_i \tag{2}$$

Interpreting Linear Regression Estimates

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- $\hat{Y}$: The predicted value of Y based on $\hat{\alpha}$ and $\hat{\beta}$:

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- Residual: the difference between observed Y and $\hat{Y}$:

$$\hat{\epsilon} = Y - \hat{Y}$$

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What would a "null line" look like?

How do we know if X predicts Y?

T-test statistic:

$$t = \frac{\hat{\beta} - \beta_{null}}{SE(\hat{\beta})} = \frac{\hat{\beta}}{SE(\hat{\beta})}$$

If we reject the null (using p-values or confidence intervals or the critical value approach), we conclude there is a statistically significant relationship between X and Y.

In other words, if we reject the null, what are we saying about β?

R Output

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  22.6838     8.2146   2.761 0.008185 **
democrat      1.0316     0.2517   4.099 0.000163 ***
---
Signif. codes:  0 *** 0.001 ** 0.01 * 0.05 . 0.1   1

Residual standard error: 12.94 on 47 degrees of freedc
Multiple R-squared:  0.2634,Adjusted R-squared:  0.247
F-statistic:  16.8 on 1 and 47 DF,  p-value: 0.0001629
```

Table: Regression Results

	Dependent variable:
	% Dems in Leg.
% Democrat	1.032*
	(0.252)
Constant	22.684*
	(8.215)
Observations	49
R^2	0.263
Note:	*p<0.05

Calculating Regression Parameters

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} \tag{3}$$

$$\hat{\beta} = \frac{\sum[(x_i - \bar{x})(y_i - \bar{y})]}{\sum(x_i - \bar{x})^2} \tag{4}$$

You must know $\hat{\beta}$ before you can calculate $\hat{\alpha}$

Y	**X**
Family Income (in thousands)	Years of Education
30	10
40	12
40	15
50	12
50	16
65	16
65	21
80	19
100	16
100	20
$\bar{Y} = 62$	$\bar{X} = 15.7$

$$
\begin{aligned}
\sum\left[(x-\bar{x})(y-\bar{y})\right] = \quad & (10-15.7)(30-62) \\
& +(12-15.7)(40-62) \\
& +(15-15.7)(40-62) \\
& +(12-15.7)(50-62) \\
& +(16-15.7)(50-62) \\
& +(16-15.7)(65-62) \\
& +(21-15.7)(65-62) \\
& +(19-15.7)(80-62) \\
& +(16-15.7)(100-62) \\
& +(20-15.7)(100-62) \\
& = 571
\end{aligned}
$$

$$
\begin{aligned}
\sum(x-\bar{x})^2 &= (10-15.7)^2 \\
&+(12-15.7)^2 \\
&+(15-15.7)^2 \\
&+(12-15.7)^2 \\
&+(16-15.7)^2 \\
&+(16-15.7)^2 \\
&+(21-15.7)^2 \\
&+(19-15.7)^2 \\
&+(16-15.7)^2 \\
&+(20-15.7)^2 \\
&= 118.1
\end{aligned}
$$

$$\hat{\beta} = \frac{\sum\left[(x-\bar{x})(y-\bar{y})\right]}{\sum(x-\bar{x})^2} = \frac{571}{118.1} = 4.83$$

$$\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x} = 62 - 4.83 \times 15.7 = -13.83$$

What is our regression equation?

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What is our regression equation?

$$Y = -13.83 + 4.83X$$

$$\text{Income} = -13.83 + 4.83(\text{Education})$$